NETWORKING

IPv4 vs IPv6 Visual Guide

Running out, then starting over — address math, headers, and transition.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/5

Address Space Math

IPv4: 2^32

4.3 billion■addresses■~1 per person■EXHAUSTED

IPv6: 2^128

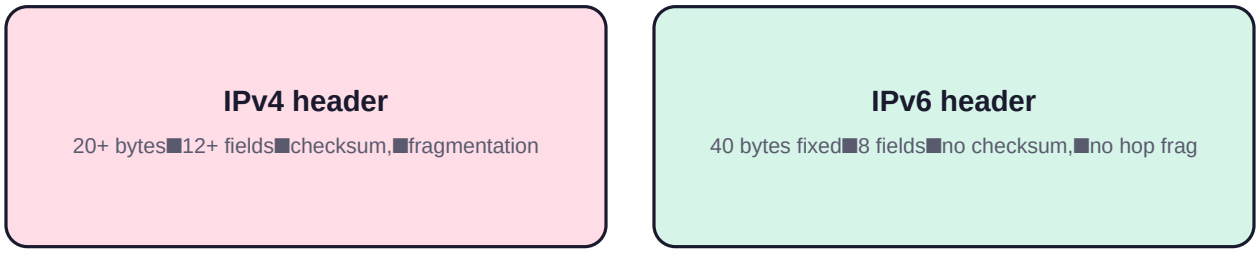
340 undecillion■~5×10^28 per person■effectively infinite

- IPv4's 32 bits seemed vast in 1981 — NAT and address markets are the expensive workaround.
- IPv6 restores the end-to-end model: every device CAN have a globally unique address again.
- Writing IPv6: 2001:db8::1 — '::' compresses one run of zero groups. /64 per LAN is the norm.

STEP 2/5

Header Simplification

IPv6 didn't just widen addresses — it cleaned house.

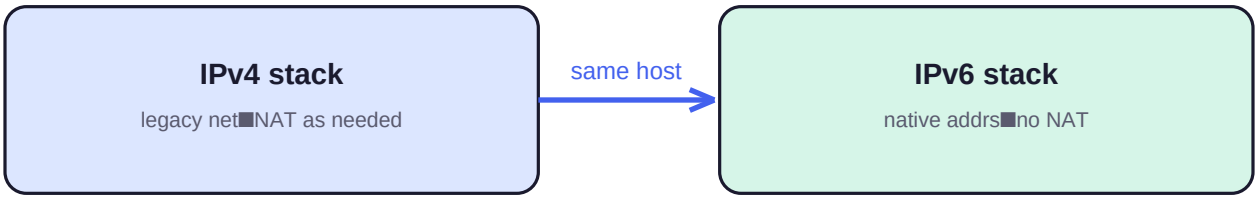


- Dropped: header checksum (links already check), in-path fragmentation (sender handles via PMTU).
- Fixed 40-byte header: routers parse faster — no variable-length options in the fast path.
- Extension headers chain optional features (routing, IPsec) without complicating the base header.

STEP 3/5

Dual-Stack: Running Both

The pragmatic transition — every host speaks two protocols.



- DNS returns both A and AAAA records; Happy Eyeballs races them and picks the working winner.
- Dual-stack doubles firewall rules, monitoring, and troubleshooting surface — budget the ops cost.
- Most cloud and mobile networks are dual-stack today; IPv6-only is growing (with NAT64 for legacy).

STEP 4/5

Transition Mechanisms

Bridges for the long migration.

NAT64 + DNS64

IPv6-only clients reach IPv4 servers: DNS synthesizes AAAA, NAT64 translates. T-Mobile US runs this.

6to4 / Teredo

Tunneling IPv6 over IPv4 — legacy transition hacks, largely retired.

464XLAT

Double translation for IPv6-only mobile networks with IPv4 apps.

There is no flag day: IPv4 and IPv6 will coexist for decades — design for both.

STEP 5/5

IPv6 Addressing in Practice

What you'll actually configure.

- SLAAC: hosts self-configure from router advertisements — no DHCP needed for addresses (DHCPv6 still gives DNS).
- Link-local (fe80::/10): every interface has one — neighbor discovery (ARP's replacement) uses it.
- Privacy extensions: hosts rotate temporary addresses so the interface ID doesn't track them.
- ULA (fc00::/7): the IPv6 'private' range — for internal use, not globally routed.

Firewall IPv6 explicitly: 'no NAT = no protection' is a myth, but an unfiltered v6 path is a real hole. Filter both stacks.

STEP ✓I✓

Key Takeaways

IPv4 vs IPv6 — Remember

- IPv4: 4.3B addresses, exhausted — NAT is the workaround. IPv6: 2^128, effectively infinite.
- IPv6 headers are fixed 40 bytes, 8 fields — no checksum, no in-path fragmentation.
- Dual-stack runs both; Happy Eyeballs picks the working protocol per connection.
- NAT64/DNS64 bridges IPv6-only clients to the IPv4 internet.
- SLAAC autoconfigures addresses; firewall the v6 path as carefully as v4.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev